Exp16:Write the python program to implement Feed forward neural Network

```
import numpy as np

# Input data
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])

# Weights and biases
W1 = np.random.rand(2, 3)
b1 = np.zeros((1, 3))
W2 = np.random.rand(3, 1)
b2 = np.zeros((1, 1))

# Activation function
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

# Feed forward process
hidden = sigmoid(np.dot(X, W1) + b1)
output = sigmoid(np.dot(hidden, W2) + b2)

print("Input:")
print(X)
print("Output:")
print(output)
```

Output

```
Input:
[[0 0]
 [0 1]
 [1 0]
 [1 1]]
Output:
[[0.75751355]
 [0.80352166]
 [0.79431654]
 [0.83233712]]
```